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Computer Evaluation of Titrations by Gran's Method

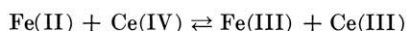
An analytical chemistry experiment

Computer evaluation of experimental data is of considerable interest in undergraduate chemistry instruction. An experiment which is readily adapted to computer techniques is the evaluation of potentiometric titration data by Gran's method of analysis. Gran's method is applicable to potentiometric acid-base, redox, precipitation, and complex formation titrations.¹ Experimental potentiometric acid-base titrations² and fluoride ion electrode titrations³ have been performed successfully by this method. The advantages of Gran's method of analysis are simplicity of measurements and calculations, versatility, and precision.²

A computer program useful for analytical chemistry labs has been developed for the evaluation of potentiometric redox titration data by Gran's method. Subroutine programs are included for least squares analyses of data and for construction of Gran plots.

Theory

The redox titration system to be considered is



The potential E (in millivolts) at 25°C of the indicator electrode against the normal hydrogen electrode for the system before the equivalence point can be determined from the Nernst equation for the Fe(II)-Fe(III) half-reaction

$$E_{\text{Fe(III),Fe(II)}} = E^{\circ'}_{\text{Fe(III),Fe(II)}} - 59.16 \log ([\text{Fe(II)}]/[\text{Fe(III)}]) \quad (1)$$

in which $E^{\circ'}$ is the formal potential for the Fe(III)-Fe(II) couple. Into the Nernst equation substitute

$$[\text{Fe(II)}] = [(V_e - V)/V_T]N \quad (2)$$

and

$$[\text{Fe(III)}] = VN/V_T \quad (3)$$

in which V_e is the volume of Ce(IV) solution to reach the end point; V the volume of Ce(IV) solution added at any point; V_T the total volume = original volume of Fe(II) solution + volume of Ce(IV) solution added; N the normality of Ce(IV) solution.

After substitution and rearrangement

$$(E - E^{\circ'})/59.16 = [\log V/(V_e - V)] \quad (4)$$

Rewriting eqn. (4) in exponential form and multiplying by $(V_e - V)/10^{E/59.16}$ leads to

$$V \cdot 10^{-E/59.16} = -V \cdot 10^{-E^{\circ'}/59.16} + V_e \cdot 10^{-E^{\circ'}/59.16} \quad (5)$$

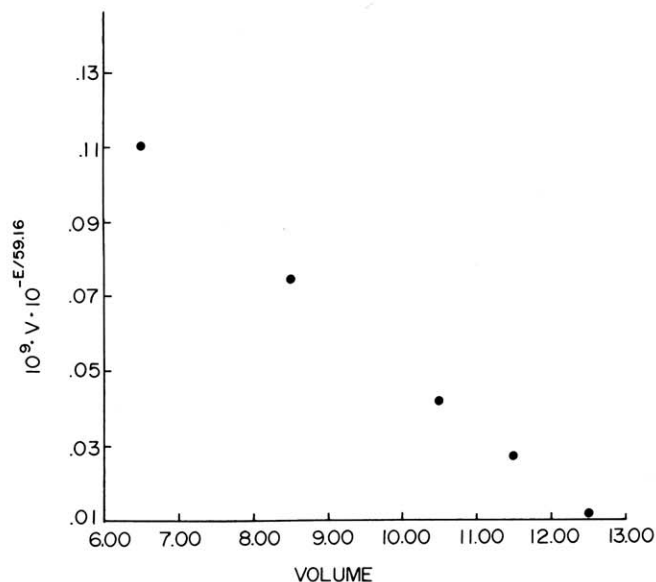
A plot of $V \cdot 10^{-E/59.16}$ versus V is a straight line of slope $-10^{-E^{\circ'}/59.16}$ and intercept $V_e \cdot 10^{-E^{\circ'}/59.16}$. V_e is determined from the intercept; $E^{\circ'}$, from either the slope or the intercept.

Experimental Procedures

The experimental procedures were similar to those indicated in many analytical chemistry textbooks. A 50-ml aliquot of a 0.03 *N* ferrous sulfate solution was titrated potentiometrically to the end point with a 0.10 *N* ceric ammonium sulfate solution. The titration, performed in sulfuric acid solution, was followed with a Sargent Model LS pH-millivolt meter equipped with a platinum-saturated calomel electrode pair. A 20% titanium trichloride solution was added at the beginning of the titration to reduce any Fe(III) to Fe(II).

Computer Program

A program⁴ written in FORTRAN IV for an IBM 360/65 computer was used to evaluate the Fe(II)-Ce(IV) titration data. The computer data input includes an identifier for each set of data, the number of points within a data set, the total volume of titrant in milliliters, the volume of titrant needed for excess titanium (determined potentiometrically), and the



Gran plot of a potentiometric Fe(II)-Ce(IV) titration.

¹ GRAN, G., *The Analyst*, **77**, 661 (1952).

² ROSSOTTI, F. J. C., AND ROSSOTTI, H., *J. CHEM. EDUC.*, **42**, 375 (1965).

³ LIBERTI, A., AND MASCINI, M., *Anal. Chem.*, **41**, 676 (1969).

⁴ A copy of the computer program is available upon request to JAC.

Table 1. A Typical Computer Printout From Evaluation of a Potentiometric Fe(II)-Ce(IV) Titration by Gran's Method of Analysis

<i>E</i>	Titration of Fe(II) with Ce(IV) Titanium End Point = 4.50 ml		<i>Y</i>
	<i>X</i>	<i>V</i>	
635.0	11.00	6.50	1.200E-10
651.0	13.00	8.50	8.421E-11
669.0	15.00	10.50	5.163E-11
680.0	16.00	11.50	3.685E-11
696.0	17.00	12.50	2.149E-11
Slope = -1.639E-11		Intercept = 2.251E-10	
Std. Dev. of a Single <i>Y</i> = 1.636E-12			
Std. Dev. of Slope = 3.397E-13			
Std. Dev. of Intercept = 3.441E-12			
End Point = 13.73 ml <i>E</i> ' of Fe(III)-Fe(II) = 638.1 mV			

Table 2. Results of Potentiometric Fe(II)-Ce(IV) Titrations by Gran's Method of Analysis

Titration	1	2	3	4
Experimental End Point [Ce(IV) soln., ml]				
Gran's Method				
Entire Titration	13.99	13.86	14.24	14.20
Last 50% of Titration	13.93	13.73	14.02	13.98
Potentiometric Method	13.90	13.75	14.05	13.95
Fe(III)-Fe(II) Formal Potential	648	640	641	646
in H ₂ SO ₄ Solution (mV)	647	638	639	641

potential in millivolts measured against a saturated calomel electrode. The measured emf is converted by a program statement to a potential against the normal hydrogen electrode. Each student readily can punch his data cards, or a laboratory assistant can prepare the cards and submit the program for an entire class. The program calculates the end point of the titration and the formal potential of Fe(III)-Fe(II) couple by least squares analysis. A point rejection routine also is included in the program. Through the introduction of

a short subroutine in the program, the weight or percent composition of a sample could be calculated. Graphs of eqn. (5) are produced by the line printer and the Cal Comp Digital Plotter. The figure illustrates the Gran plot of a selected titration. A computer output for the selected titration is indicated in Table 1 where *E* is the potential in millivolts against a normal hydrogen electrode, *X* is the total volume of cerium solution added, *V* is the volume of cerium solution needed for the titration of iron, and *Y* is the function $V \cdot 10^{-E/59.16}$.

Results and Discussion

The results of several potentiometric Fe(II)-Ce(IV) titrations by Gran's method of analysis are summarized in Table 2. Indicated are the end point volumes determined both potentiometrically and by Gran's method and the calculated formal potentials in sulfuric acid solution of the Fe(III)-Fe(II) couple. Gran plots were obtained for each titration from evaluation of data for the entire titration and for the last 50% of the titration. Nonlinear plots resulted from use of data for the entire titration. Curvature at the beginning of the titration was similar to that obtained in Gran plots of acid-base systems.² This increased the value of the intercept and therefore led to high values of the end point volume. Plots constructed for the last half of the titration gave end point volumes in excellent agreement with those determined potentiometrically. A constant formal potential for the Fe(III)-Fe(II) couple of 0.64 V was obtained although no attempt was made to hold the ionic strength or the temperature constant. This value compares favorably with the Fe(III)-Fe(II) formal potential of 0.68 V in 0.1 M, 0.5 M, and 4 M H₂SO₄.⁵

⁵ LAITINEN, H. A., "Chemical Analysis, An Advanced Text and Reference," McGraw-Hill, New York, 1960, p. 283.